THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · NCERT-TYPE CHAPTER TEST SET 3

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. In a number pyramid each block is the sum of the two below it. The bottom row is 3, 3, 2, 5. The top block is: 1

(a) 23 **(b)** 18 **(c)** 26 **(d)** 13

2. In a calendar the nine dates in a 3×3 block have the middle date 19. Their sum is: 1

(a) 171 **(b)** 57 **(c)** 162 **(d)** 180

3. The value of $25 \times 25 - 24 \times 26$ is: 1

(a) 2 **(b)** 1 **(c)** 0 **(d)** 25

4. For any n, $(n+4)^2 - (n-4)^2$ equals: 1

(a) 32 **(b)** 64 **(c)** $16n$ **(d)** $8n$

5. Think of a number n. Add 5, multiply by 6, then subtract 30. The result is: 1

(a) $n + 5$ **(b)** $6n + 5$ **(c)** $6n$ **(d)** $5n$

6. The sum of 7 consecutive integers starting with n is: 1

(a) $n + 21$ **(b)** $7n$ **(c)** $7n + 21$ **(d)** $7n + 22$

7. In a number pyramid each block is the sum of the two below it. The bottom row is 6, 2, 4, 1. The top block is: 1

(a) 19 **(b)** 26 **(c)** 25 **(d)** 13

8. In a calendar the nine dates in a 3×3 block have the middle date 23. Their sum is: 1

(a) 207 **(b)** 69 **(c)** 216 **(d)** 198

9. The value of $40 \times 40 - 39 \times 41$ is: 1

(a) 2 **(b)** 40 **(c)** 1 **(d)** 0

10. For any n, $(n+5)^2 - (n-5)^2$ equals: 1

(a) $10n$ **(b)** 50 **(c)** 100 **(d)** $20n$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the sum of the 6 consecutive integers 20, 21, ..., 25 using the expression $6n + 15$. 2

12. Show that the sum of 27 and its reverse is divisible by 11. 2

13. Think of a number n, add 6, multiply by 6, subtract 36, divide by 6. Show that the answer is n. 2

14. Find the sum of the 4 consecutive integers 33, 34, ..., 36 using the expression $4n + 6$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. In a number pyramid with bottom row 3, 1, 5, 2, each block is the sum of the two below it. Find all the blocks. 3

16. Show that the difference of 95 and its reverse 59 is a multiple of 9. 3

17. A 3×3 block of calendar dates has middle date 22. Write the nine dates and show that their sum is 9 times the middle date. 3

OR

In a number pyramid with bottom row 6, 2, 2, 3, each block is the sum of the two below it. Find all the blocks. 3

18. Show that the difference of 82 and its reverse 28 is a multiple of 9. 3

OR

A 3×3 block of calendar dates has middle date 24. Write the nine dates and show that their sum is 9 times the middle date. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Show that the sum of 7 consecutive integers starting with n is $7n + 21$ and verify it for $n = 15$. 5

OR

Think of a number n. Double it, add 12, halve the result and subtract n. Show algebraically that the final answer is always 6. Verify for $n = 20$. 5

20. Show that the sum of 8 consecutive integers starting with n is $8n + 28$ and verify it for $n = 30$. 5

OR

Think of a number n. Double it, add 18, halve the result and subtract n. Show algebraically that the final answer is always 9. Verify for $n = 6$. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND